

IMO PHASE II LEVEL 2

PROBABILISTIC METHOD(With Solution)

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1 Linearity of Expectation

Definition 1.1 (Random Variable)

A **random variable** is a variable whose value is unknown or a function that assigns values to each of an experiment's outcomes.

Definition 1.2 (Expected Value)

The **expected value** is the "weighted average" of a random variable X :

$$\mathbb{E}[X] := \sum_x \mathbb{P}(X = x) \cdot x.$$

Example 1.3

There are n people, each of who has a name tag. We shuffle the name tags and randomly give each person one of the name tags. Let S be the number of people who receive their own name tag. Prove that the expected value of S is 1. (i.e. S independent of n)

Proof. We can list all the $n!$ permutations and count the number of cases where the i -th person get his own name tag. We can easily see that there are $(n-1)!$ cases where the i -th person get his own tag. Hence we have $\mathbb{E}[S] = \frac{1}{n!} ((n-1)! * n) = 1$. $\square$

Theorem 1.4 (Linearity of Expectation)

Given any random variables $X_1, X_2, \dots, X_n$, we always have

$$\mathbb{E}[X_1 + X_2 + \dots + X_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \dots + \mathbb{E}[X_n].$$

Remark. It is true even if X_i s are dependent of each other.

Proof.

$$\begin{aligned} \mathbb{E}[X + Y] &= \sum_x \sum_y [(x + y) \cdot P(X = x, Y = y)] \\ &= \sum_x x \sum_y P(X = x, Y = y) + \sum_y y \sum_x P(X = x, Y = y) \\ &= \sum_x x \cdot P(X = x) + \sum_y y \cdot P(Y = y) \\ &= \mathbb{E}[X] + \mathbb{E}[Y]. \end{aligned}$$

□

Example 1.5

At a nursery, 2006 babies sit in a circle. Suddenly, each baby randomly pokes either the baby to its left or to its right. What is the expected value of the number of unpoked babies?

Proof. Number the babies $1, 2, \dots, 2006$. Define

$$X_i := \begin{cases} 1, & \text{if baby } i \text{ is unpoked} \\ 0, & \text{otherwise.} \end{cases}$$

We seek $\mathbb{E}[X_1 + X_2 + \dots + X_{2006}]$. Note that any particular baby has probably $(\frac{1}{2})^2 = \frac{1}{4}$ of being unpoked. Hence $\mathbb{E}[X_i] = \frac{1}{4}$ for each i and hence

$$\mathbb{E}[X_1 + \dots + X_2 + \dots + X_{2006}] = \sum_{i=1}^{2006} \mathbb{E}[X_i] = 2006 \cdot \frac{1}{4} = \frac{1003}{2}.$$

□

Usage 1.6 (Use Expected Value to Show Existence)

Suppose we know the average score of a test is 12.1, then there exists a student who got at least 13 points, and a contestant who got at most 12 points.

Remark. It is similar in spirit to the pigeonhole principle.

Example 1.7

Prove that any subgraph of $K_{n,n}$ with at least $n^2 - n + 1$ edges has a perfect matching.

Proof. We randomly pair off one set of points with the other (whether there is an actual edge or not), and define the score of such a pairing be the number of pairs which are actually connected by an edge. Number the pairs by $1, 2, \dots, n$. Define

$$X_i := \begin{cases} 1 & \text{if the } i\text{th pair is connected by an edge} \\ 0 & \text{otherwise.} \end{cases}$$

Hence the score of a config $X = X_1 + \dots + X_n$. For any pair of points, they are connected with probability at least $\frac{n^2 - n + 1}{n^2}$. Hence $\mathbb{E}[X_i] = \frac{n^2 - n + 1}{n^2}$, and hence

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n] = n \cdot \mathbb{E}[X_1] \\ &= \frac{n^2 - n + 1}{n} \\ &= n - 1 + \frac{1}{n}. \end{aligned}$$

Since X takes only integer values, there exists a config with $X = n$.

□

Theorem 1.8 (Jenson's Inequality)

If f is a convex function and $t \in [0, 1]$, we have

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2).$$

In context of probability theory, if X is a random variable and φ is a convex function, then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

2 Problems for Discussion

Problem 1. (IMC 2002 day2 q2) 200 students participated in a math contest. They had 6 problems to solve. Each problem was correctly solved by at least 120 participants. Prove that there must be 2 participants such that every problem was solved by at least one of these two students.

Ans. Pick two contestants randomly. Let X_i be the indicator that both contestants miss problem i , so each $\mathbb{E}[X_i] < \left(\frac{80}{200}\right)^2 = 4/25$, and their expected number of both-missed problems is $24/25 < 1$.

Problem 2. (Russia 1996) In the Duma there are 1600 delegates, who have formed 16000 committees of 80 persons each. Prove that one can find two committees having no fewer than four common members.

Ans. Consider a random couple of committees, and let X be the number of delegates that join both of them. Consider a_d the number of committees that include delegate d . As the average person is on $16000 \cdot 80 / 1600 = 800$, by Jensen's inequality the expected value of X is

$$\mathbb{E}[X] = \sum_d \frac{\binom{a_d}{2}}{\binom{16000}{2}} \geq \frac{1600 \left(\frac{\sum a_d}{1600} \right)}{\binom{16000}{2}} = \frac{1600 \left(\frac{16000 \cdot 80}{1600} \right)}{\binom{16000}{2}} = \frac{1600 \binom{800}{2}}{\binom{16000}{2}} > 3,$$

and the conclusion follows.

Problem 3. (BAMO 2004) Consider n real numbers, not all zero, with sum zero. Prove that one can label the numbers as $a_1, a_2, \dots, a_n$ such that $a_1 a_2 + a_2 a_3 + \dots + a_n a_1 < 0$.

Ans. Show that a random permutations has expected value at most 0. By summing all the equations with $n!$ permutations. We can get $\sum -a_i^2$. Since they are not all 0 we have expected value < 0 , hence the sum.

$$\mathbb{E}[a_i a_{i+1}] = \mathbb{E}[a_i \mathbb{E}[a_{i+1} | a_i]] = \mathbb{E}\left[a_i \left(\frac{-a_i}{n-1} \right)\right] = -\mathbb{E}\left[\frac{a_i^2}{n-1}\right] < 0.$$

Problem 4. (IMOSL 1999 C4) Let A be any set of n residues mod n^2 . Show that there is a set B of n residues mod n^2 that at least half of the residues mod n^2 can be written as $a + b$ with a in A and b in B .

Ans. Select elements of $B = \{b_1, \dots, b_n\}$ uniformly randomly (we'll even allow repetitions for simplicity). For each $r \pmod{N^2}$, and each $i, j \in 1, \dots, n$, by considering the number of choices of A , we have

$$\mathbb{P}(r \notin A + b_i) = \mathbb{P}(b_i \notin A - r) = \frac{N^2 - N}{N^2} = 1 - \frac{1}{N}.$$

Hence $\mathbb{P}(r \notin A + B) < \left(1 - \frac{1}{N}\right)^N < \frac{1}{e} < \frac{1}{2}$. Thus $\mathbb{E}[|A + B|] > (1 - 1/2)N^2$.

Problem 5. (Online Math Open q29) Kevin has $2^n - 1$ cookies, each labeled with a unique nonempty subset of $\{1, 2, \dots, n\}$. Each day, he chooses one cookie uniformly at random out of the cookies not yet eaten. Then, he eats that cookie, and all remaining cookies that are labeled with a subset of that cookie. Compute the expected value of the number of days that Kevin eats a cookie before all cookies are gone.

Ans. The number of days equals the number of times a cookie is chosen (rather than merely eliminated). Let C be the set of cookies chosen by the process and $S = \{1, \dots, n\}$, so

$$\mathbb{E}[\# \text{days}] = \mathbb{E}[|C|] = \sum_{A \subseteq S} \mathbb{E}[\text{ind}(A \in C)] = \sum_{A \subseteq S} \mathbb{P}(A \in C).$$

And since cookie A is uneaten iff all its supersets are also uneaten, we have

$$(A \in C) = \frac{1}{\# \{\text{supersets of } A\}} = \frac{1}{2^{n-|A|}}.$$

Hence

$$\sum_{A \subseteq S} \mathbb{P}(A \in C) = \sum_{A \subseteq S} 2^{|A|-n} = \sum_{1 \leq k \leq n} 2^k \cdot 2^{k-n} = \frac{4^{n+1} - 4}{3 \cdot 2^n}.$$

Problem 6. (EGMO 2019 P5) Let $n \geq 2$ be an integer, and let $a_1, a_2, \dots, a_n$ be positive integers. Show that there exist positive integers $b_1, b_2, \dots, b_n$ satisfying the following three conditions:

- (A) $a_i \leq b_i$ for $i = 1, 2, \dots, n$;
- (B) the remainders of $b_1, b_2, \dots, b_n$ on division by n are pairwise different; and
- (C) $b_1 + b_2 + \dots + b_n \leq n \left(\frac{n-1}{2} + \left\lfloor \frac{a_1 + a_2 + \dots + a_n}{n} \right\rfloor \right)$

Ans. Note that if $a_i \geq n$, we can replace a_i with $a_i - n$ and also b_i with $b_i - n$ without affecting the validity of the problem (and if it holds for $a_i > 0$, there is no reason for it to fail at $a_i = 0$); therefore, we may assume that each a_i is between 0 and $n - 1$.

Define a new sequence $c_i = b_i \pmod{n}$. For some a_i , we randomly select the value of the corresponding c_i ; notice that there is a $\frac{a_i}{n}$ chance that the value of c_i is less than a_i , and therefore it will be necessary that b_i is equal to $c_i + n$. We have $\sum b_i \leq \sum a_i + \sum c_i$ by LoE; notice that $\sum c_i = \frac{n(n-1)}{2}$, and by mod n , we can indeed replace $\sum a_i$ with $n \lfloor \frac{\sum a_i}{n} \rfloor$, and therefore obtaining the bound in the original problem statement.

Problem 7. (IMC 2017 q4) There are n people in a city, and each of them has exactly 1000 friends (friendship is always symmetric). Prove that it is possible to select a group S of people such that at least $\frac{n}{2017}$ persons in S have exactly two friends in S .

Ans. Let $d = 1000$ and let $0 < p < 1$. Choose the set S randomly such that each people is selected with probability p , independently from the others.

The probability that a certain person is selected for S and knows exactly two members of S is $q = \binom{d}{2} p^3 (1-p)^{d-2}$. Choose $p = 3/(d+1)$ (this is the value of p for which q is maximal); then

$$\begin{aligned} q &= \binom{d}{2} \left(\frac{3}{d+1} \right)^3 \left(\frac{d-2}{d+1} \right)^{d-2} \\ &= \frac{27d(d-1)}{2(d+1)^3} \left(1 + \frac{3}{d-2} \right)^{-(d-2)} \\ &> \frac{27d(d-1)}{2(d+1)^3} \cdot e^{-3} \\ &> \frac{1}{2017}. \end{aligned}$$

Hence $E(|S|) = nq > \frac{n}{2017}$, so there is a choice for S when $|S| > \frac{n}{2017}$.

Problem 8. (Iran TST 2008 P6) Prove that in a tournament with 799 teams, there exist 14 teams, that can be partitioned into groups in a way that all of the teams in the first group have won all of the teams in the second group.

Ans. We are going to prove we can partition the group of 14 into two groups of 7.

Let D_k be the set of teams which defeat the k -th team (here $1 \leq k \leq 799$), and $d_k = |D_k|$. Select $A = \{a_1, \dots, a_7\}$ randomly, so $\mathbb{P}(A \subseteq D_k) = \binom{d_k}{7} / \binom{799}{7}$. Let N be the number of teams dominated by A , we have $\mathbb{E}[N] = \sum_k \binom{d_k}{7} / \binom{799}{7}$. Since $\binom{x}{7}$ is convex, and the average value of d_k is $798/2 = 398$, by Jensen's we have

$$\mathbb{E}[N] \geq 799 \cdot \binom{398}{7} / \binom{799}{7} > 799 \cdot \left(\frac{1}{2} \right)^7 > 6.$$

Hence sometimes $N \geq 7$.

Problem 9. (Caro-Wei Theorem) Consider a graph G with vertex set V . Prove that one can find an independent set of V (pairwise non-adjacent) with size at least

$$\sum_{v \in V} \frac{1}{\deg v + 1}.$$

Remark. By applying Jensen's inequality, our independent set has size at least $\frac{n}{d+1}$, where d is the average degree. This result is called Turan's Theorem.

Ans. Randomly order the vertices $\{v_1, \dots, v_n\}$ of G , and take W to be the subset of those v_i which occur before all their neighbors. Then

$$\mathbb{E}[|W|] = \sum_i \mathbb{P}(v_i \in W) = \sum_i \frac{1}{\deg(v) + 1}.$$